Contents available at ScienceDirect

Diabetes Research and Clinical Practice

journal homepage: www.elsevier.com/locate/diabres

International Diabetes Federation

Risk of progression to diabetes from prediabetes defined by HbA1c or fasting plasma glucose criteria in Koreans

Chul-Hee Kim [a], *Hong-Kyu Kim* [b,*], *Eun-Hee Kim* [b], *Sung-Jin Bae* [b], *Jaewon Choe* [b], *Joong-Yeol Park* [c]

[a] *Division of Endocrinology & Metabolism, Department of Internal Medicine, Soonchunhyang University College of Medicine, Bucheon, Republic of Korea*

[b] *Health Screening & Promotion Center, Asan Medical Center, Seoul, Republic of Korea*

[c] *Division of Endocrinology & Metabolism, Department of Internal Medicine, University of Ulsan College of Medicine, Seoul, Republic of Korea*

ARTICLE INFO

Article history:
Received 1 October 2015
Received in revised form
23 March 2016
Accepted 6 June 2016
Available online 18 June 2016

Keywords:
Prediabetes
HbA1c
Fasting plasma glucose
Diabetes risk

ABSTRACT

Aims: To examine the abilities of HbA1c and fasting plasma glucose (FPG) criteria predicting 5-year progression rate to diabetes in Korean adults with prediabetes.

Methods: Participants included 17,971 Koreans (aged 20–79 years) who underwent routine medical check-ups at a mean interval of 5.2 years (3.1–6.7 years). Prediabetes was defined as FPG 5.6–6.9 mmol/l or HbA1c 5.7–6.4% (39–46 mmol/mol). Incident diabetes was defined as FPG ⩾ 7.0 mmol/l, HbA1c ⩾ 6.5% (48 mmol/mol), or initiation of antidiabetic medications.

Results: At baseline, the prevalence of prediabetes was 30.6% ($n = 5495$) by FPG and 20.4% ($n = 3664$) by HbA1c criteria. The 5-year progression rate to diabetes was significantly higher in prediabetes identified by HbA1c than by FPG tests (14.7% vs. 10.4%, $P < 0.001$). Of individuals diagnosed with prediabetes by only one test, those by HbA1c alone had a higher risk of progression to diabetes than those diagnosed by FPG alone (6.0% vs. 3.9%, $P < 0.001$). Receiver operating characteristic curve analysis showed that area under the curve was greater for HbA1c (0.855, 95% CI 0.840–0.870) than for FPG (0.830, 0.813–0.846) ($P = 0.016$). After adjustment for conventional risk factors, the odds ratio (OR) of developing diabetes was higher in participants with prediabetes identified by HbA1c (OR 9.91, 8.24–11.9) than by FPG (OR 7.29, 5.97–8.89) ($P = 0.026$).

Conclusions: Although fewer individuals with prediabetes were identified by HbA1c than by FPG criteria, the ability to predict progression to diabetes was stronger for HbA1c than for FPG in Koreans.

© 2016 Elsevier Ireland Ltd. All rights reserved.

1. Introduction

Prediabetes is associated with significant risks of developing type 2 diabetes mellitus (T2DM) and cardiovascular comorbidities [1]. Early screening and diagnosis of prediabetes, along with subsequent preventive measures, are vital in preventing or delaying progression to T2DM. The American Diabetes Association (ADA) has utilized hemoglobin A1c

* Corresponding author at: Health Screening & Promotion Center, Asan Medical Center, 88 Olympic-ro 43-gil, Songpa-gu, Seoul 05505, Republic of Korea. Fax: +82 2 3010 4964.
E-mail address: hkkim0801@amc.seoul.kr (H.-K. Kim).

http://dx.doi.org/10.1016/j.diabres.2016.06.009
0168-8227/© 2016 Elsevier Ireland Ltd. All rights reserved.

(HbA1c) concentration to diagnose diabetes mellitus, categorizing individuals with HbA1c levels above the "normal" range but below the diagnostic cutoff point for diabetes as having prediabetes [2]. Although both HbA1c and fasting plasma glucose (FPG) concentrations have been recommended as screening tests for prediabetes, with both increased FPG and HbA1c being well-established risk factors for progression to diabetes, the better predictor of diabetes development has not been determined yet. In addition, there has been some controversy about the criteria for defining prediabetes. The WHO defined impaired fasting glycemia as FPG 6.1–6.9 mmol/l [3] while the ADA defined it as 5.6–6.9 mmol/l. For the HbA1c criteria, the International Expert Committee recommended using 6.0–6.4% [42–46 mmol/mol] [4], conflicting with the ADA definition of 5.7–6.4% [39–46 mmol/mol].

Epidemiological studies [5–7] show significant discordance between HbA1c and glucose-based tests results in defining high risks for diabetes. Racial and ethnic differences in HbA1c have been described that do not appear to be explained by differences in glycemia [8–10]. Therefore, the degree of agreement between HbA1c and FPG-based criteria in the diagnosis of prediabetes was poor, and varied among ethnic groups and populations or according to degree of obesity [8,10,11]. We also observed a significant discordance between FPG- and HbA1c-based criteria in diagnosing diabetes mellitus and prediabetes among Koreans, with the groups diagnosed by each criterion differing in age, gender, blood pressure, and insulin resistance [12–14]. Although HbA1c alone has been found to identify fewer individuals with prediabetes than FPG criteria alone, about 20% more could be detected by the addition of HbA1c to FPG criteria in a Korean cohort study [13]. However, most previous studies were cross-sectional in design, and the differences in actual rate of progression to diabetes remain to be elucidated.

Several longitudinal studies [15–18] have compared the ability of HbA1c and FPG concentrations to predict future risks of diabetes, but those studies yielded conflicting results. For example, a study of older adults in Finland reported that similar proportions of participants with impaired glucose tolerance (IGT), elevated HbA1c, and impaired fasting glucose (IFG) developed diabetes [15]. By contrast, a study of older adults in the US reported that the adjusted odds ratio (OR) of diabetes was higher in those with elevated HbA1c than with IFG [18]. Ethnic differences have been observed in the rate of conversion from prediabetes to type 2 diabetes, with more rapid conversion in populations of South-Asian and African origin than of European origin [19]. Less is known, however, about East-Asian populations; to our knowledge, there have been only two published reports to date from a Japanese cohort [16,17]. This study assessed the 5-year progression to diabetes in a large cohort of Korean adults identified as having prediabetes by HbA1c and FPG criteria. In addition, the effects of age, gender, and body mass index (BMI) on the performance of these tests were evaluated.

2. Participants and methods

This study retrospectively assessed clinical and laboratory data obtained from 19,356 participants (aged 20–79 years) who visited the Health Screening and Promotion Center at Asan Medical Center (Seoul, Korea) for medical examinations in 2007–2008 and underwent follow-up examinations in 2012–2013. Individuals with a history of physician-diagnosed diabetes mellitus, those taking anti-hyperglycemic medications, and those with FPG $\geqslant$ 7.0 mmol/l or HbA1c $\geqslant$ 6.5% (48 mmol/mol) at baseline (n = 1296) were excluded. After excluding additional 89 individuals with missing follow-up data on glycemic status, data were analyzed for 17,971 participants, 10,965 (61.0%) men and 7006 (39.0%) women. Prediabetes was defined as FPG 5.6–6.9 mmol/l or HbA1c 5.7–6.4% (39–46 mmol/mol). We subdivided individuals with prediabetes by level of glycemia, with FPG concentrations 5.6–6.0 mmol/l or 6.1–6.9 mmol/l and HbA1c 5.7–5.9% (39–41 mmol/mol) or 6.0–6.4% (42–46 mmol/mol). The study protocol was in accordance with the Declaration of Helsinki (as revised in Seoul, 2008) and was approved by the institutional review board of Asan Medical Center. All participants provided written informed consent.

At the initial visit, each participant filled out a questionnaire on alcohol intake, smoking habits, physical activity, medical history, family history of diabetes, and use of regular medications. Anthropometric and blood pressure measurements were taken by trained personnel. Blood pressure was measured on the right arm in a sitting position after a 5-min rest using an electric sphygmomanometer. Blood samples were obtained in the morning after an overnight fast of at least 8 h. Plasma glucose was measured by the hexokinase method using an auto-analyzer (TBA-200FR, Toshiba, Tokyo, Japan). Total cholesterol, HDL-cholesterol, and triglycerides levels were also measured using an auto-analyzer (Toshiba). HbA1c was measured using ion-exchange high-performance liquid chromatography via an automated analyzer (Variant II™, Bio-Rad Laboratories, Hercules, CA, USA), which was aligned to the Diabetes Control and Complications Trial reference standard and certified by the National Glycohemoglobin Standardization Program.

Incident diabetes was defined as FPG $\geqslant$ 7.0 mmol/l, HbA1c $\geqslant$ 6.5% (48 mmol/mol), or the initiation of anti-hyperglycemic medications during the follow-up period.

Statistical analyses were performed using SPSS 21.0 software (IBM, Armonk, NY, USA). Baseline characteristics of each group were compared using unpaired Student's t-test or chi-square tests. Multivariate logistic regression analyses were used to obtain odds ratios (ORs) for the development of diabetes after adjusting for clinical and biochemical variables that showed significant association in univariate analysis and clinical relevance. Areas under receiver operating characteristic (ROC) curves were evaluated to compare the ability of FPG and HbA1c to predict the development of diabetes using the method of DeLong [20]. P values less than 0.05 were considered statistically significant.

3. Results

Of the 17,971 participants who did not have diabetes at baseline, 5495 (30.6%) had prediabetes by FPG, 3664 (20.1%) by HbA1c, and 7208 (40.1%) by combined criteria. The baseline characteristics of the participants according to glycemic

status are shown in Table 1. Prediabetic individuals who met HbA1c criteria only were significantly older and more likely to be women, whereas those who met FPG criteria only had a higher BMI, waist circumference, and systolic and diastolic blood pressure.

The mean follow-up period in the study cohort was 5.2 years (range 3.1–6.7 years), during which 723 participants (4.0%) developed diabetes. Among the newly diagnosed diabetic patients, 607 (84%) could be diagnosed by HbA1c criteria, 500 (69%) by FPG criteria, and 226 (31%) by medication. Only 0.4% of participants who had normoglycemia (FPG < 5.6 mmol/l and HbA1c < 5.7% [39 mmol/mol]) at baseline developed diabetes, compared with 9.4% of those with prediabetes by any criterion. The 5-year rate of progression to diabetes was significantly higher in participants identified as having prediabetes by HbA1c than by FPG criteria (14.7% vs. 10.4%, $P < 0.001$). This trend was consistent in all subgroups divided by age (<50 or >50 years), sex, BMI (<23, 23–27, or >27 kg/m^2), and family history of diabetes (data not shown). Of those individuals diagnosed with prediabetes by one criterion alone, those diagnosed by HbA1c concentration alone had a higher risk of progression to diabetes than those diagnosed by FPG concentration alone (6.0% vs. 3.9%, $P < 0.001$). Individuals diagnosed with prediabetes by both criteria had a much higher rate of progression to diabetes (22.3%).

When participants with prediabetes were subdivided by level of glycemia, the risks of progression to diabetes were substantially higher in individuals with higher (6.1–6.9 mmol/l) than lower (5.6–6.0 mmol/l) FPG concentrations, as well as in those with higher (6.0–6.4% [42–46 mmol/mol]) than lower (5.7–5.9% [39–41 mmol/mol]) HbA1c concentrations (Fig. 1). ROC curve analysis for the development of diabetes showed that the cut-off values with maximum sensitivity plus specificity were 5.7 mmol/l (102 mg/dl) for FPG (sensitivity 75.1%, specificity 78.0%) and 5.7% (39 mmol/mol) for HbA1c (sensitivity 74.4%, specificity 81.9%). The area under the curve (AUC) was greater for HbA1c (0.855, 95% CI 0.840–0.870) than for FPG (0.830, 95% CI 0.813–846) (P value for AUC difference = 0.016) (Fig. 2). AUC of ROC curve using combination of FPG and HbA1c (0.898, 95% CI 0.886–0.909) was significantly greater than those of either test alone ($P < 0.001$).

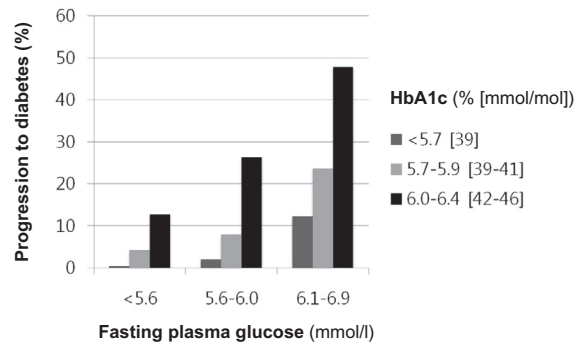

Fig. 1 – Rate of progression to diabetes over 5 years relative to fasting plasma glucose and HbA1c concentrations at baseline.

The sensitivities and specificities for prediction of diabetes by each criterion were: 79.1% and 71.5% for FPG 5.6–6.9 mmol/l; 74.4% and 81.9% for HbA1c 5.7–6.4% [39–46 mmol/mol]; 49.4% and 93.8% for FPG 6.1–6.9 mmol/l; 44.5% and 95.5% for HbA1c 6.0–6.4% [42–46 mmol/mol], respectively. Combining the two tests could markedly increase the sensitivity but decreased specificity: 93.4% and 62.1% for FPG 5.6–6.9 mmol/l plus HbA1c 5.7–6.4% [39–46 mmol/mol]; 68.7% and 90.4% for FPG 6.1–6.9 mmol/l plus HbA1c 6.0–6.4% [42–46 mmol/mol].

Multivariate logistic regression analysis showed that, after adjustments for age, sex, BMI, smoking, alcohol intake, physical activity, family history of diabetes, hypertension, serum triglycerides, and HDL-cholesterol levels, the OR for developing diabetes was significantly higher in individuals diagnosed with prediabetes by HbA1c (OR 9.91, 95% CI 8.24–11.9) than by FPG (OR 7.29, 95% CI 5.97–8.89) criteria ($P = 0.026$). Subdivision

Table 1 – Baseline characteristics of participants according to prediabetes status.

	Normo-glycemia	Prediabetes by		
		FPG only	HbA1c only	Both
n	10,763	3544	1713	1951
Age (years)	47.0 ± 8.7	49.5 ± 7.9	51.2 ± 8.5	52.1 ± 8.2
Sex (M/F, %)	55/45	76/24	53/47	74/26
BMI (kg/m^2)	23.2 ± 2.8	24.4 ± 2.7	23.9 ± 2.8	25.1 ± 2.8
Waist circumference (cm)	80.0 ± 8.6	84.5 ± 7.9	82.2 ± 8.6	86.4 ± 7.8
Systolic BP (mmHg)	114 ± 13	120 ± 13	116 ± 14	121 ± 14
Diastolic BP (mmHg)	71 ± 9	75 ± 9	72 ± 9	76 ± 9
FPG (mmol/l)	5.0 ± 0.3	5.9 ± 0.3	5.1 ± 0.3	6.0 ± 0.4
HbA1c (%, mmol/mol)	5.2 ± 0.3 (33 ± 2)	5.3 ± 0.3 (34 ± 2)	5.8 ± 0.2 (40 ± 2)	5.9 ± 0.2 (41 ± 2)
Total cholesterol (mmol/l)	4.78 ± 0.81	4.97 ± 0.85	5.04 ± 0.88	5.12 ± 0.89
Triglycerides (mmol/l)	1.26 ± 0.75	1.56 ± 0.96	1.50 ± 0.95	1.79 ± 1.19
HDL-cholesterol (mmol/l)	1.50 ± 0.37	1.43 ± 0.35	1.44 ± 0.37	1.37 ± 0.32
Regular exercise (⩾3 times/wk) (%)	42.7	46.5	45.6	46.2
Alcohol consumption (⩾2 times/wk) (%)	38.1	58.0	33.0	50.3
Family history of DM (%)	18.1	21.8	22.1	26.8
Current smoker (%)	21.9	24.0	25.1	28.0

FPG, fasting plasma glucose; BP, blood pressure; HDL, high-density lipoprotein; DM, diabetes mellitus.

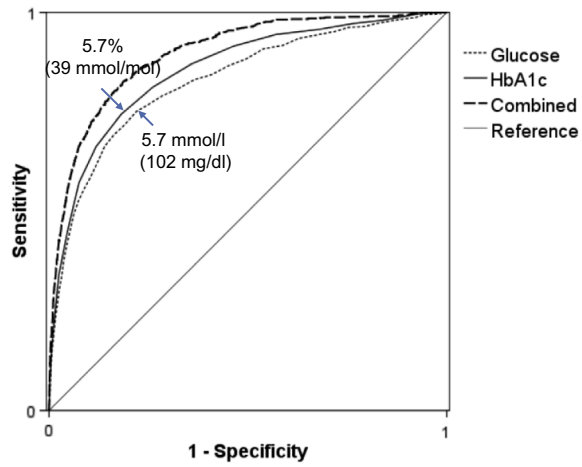

Fig. 2 – Receiver operating characteristic curve for the development of diabetes relative to baseline fasting plasma glucose and HbA1c concentrations. Arrows indicate the points of maximum sensitivity plus specificity.

of participants with prediabetes showed that the OR of progression to diabetes was higher in individuals with FPG 6.1–6.9 mmol/l than with FPG 5.6–6.0 mmol/l and in those with HbA1c 6.0–6.4% (42–46 mmol/mol) than HbA1c 5.7–5.9% (39–41 mmol/mol) (Table 2). Individuals with both higher FPG (6.1–6.9 mmol/l) and higher HbA1c (6.0–6.4% [42–46 mmol/mol]) had the highest risk of progression to diabetes (182/380, 47.9%; OR 163, 95% CI 111–238).

4. Discussion

This large scale longitudinal study showed that the 5-year rate of progression to diabetes was higher in individuals with prediabetes identified by HbA1c than by FPG criteria. In addition, the area under the ROC curve for development of diabetes was greater for HbA1c than for FPG. These results indicated HbA1c was a better predictor of progression to diabetes than FPG, although fewer individuals with prediabetes were identified using HbA1c than FPG criteria. These two criteria could independently and additively predict risk of progression to diabetes, consistent with previous studies showing that combining the two tests was better than either alone in identifying individuals with an increased risk of diabetes [17,18,21,22].

Limited number of studies to date have compared the incidence of new-onset diabetes in people with prediabetes, as determined by HbA1c and FPG criteria according to current ADA guidelines. Our results are consistent with those in older (aged 70–79 years) American adults, which showed that the OR of diabetes was higher in individuals with elevated HbA1c than those with IFG [18]. A study in Hong Kong Chinese also reported that HbA1c had a higher standardized hazard ratio and greater sensitivity and specificity than FPG in predicting the 8-year incidence of diabetes [23]. By contrast, studies in Japan [24] and Spain [25] reported that FPG and HbA1c were similarly predictive, and a study of older adults (aged 61–63 years) in Finland found that the proportion of participants with IGT, elevated HbA1c, and IFG who developed diabetes within 10 years was similar (37–38%) [15]. Many of these studies, however, included fewer participants or those within a limited age range and used different cut-offs of HbA1c for prediabetes. A large study of Japanese government employees (n = 6241) showed that ADA criteria for prediabetes using HbA1c and FPG were similar in their ability to predict progression to diabetes [16]. It is difficult to compare the predictive ability of HbA1c and FPG between studies because of differences in age of the participants, criteria for prediabetes, and duration of follow up [26]. In addition, differences in ethnicity, degree of obesity at baseline, lifestyle factors, and socio-economic status could influence the rate of progression to diabetes. The baseline characteristics of our study participants, including age, BMI, and lipid profiles, were similar to those of the Japanese study [16]. However, the Japanese cohort was a relatively homogeneous population of government employees with male predominance, while our participants had a more diverse socio-economic status and a higher proportion of women (39% vs. 25%). Our finding that HbA1c was a better predictor of progression to diabetes than FPG was unsurprising, in that HbA1c is a more stable marker of glycemia with less intra-individual variation and it reflects not only fasting but also postprandial hyperglycemia. However, it should be noted that some incorporation bias might be involved in our study since 15% more subjects were diagnosed as having diabetes by HbA1c than by FPG at follow up. Further confirmation would be needed in other populations.

The baseline prevalences of prediabetes (30.6% by FPG, 20.4% by HbA1c, and 40.1% by combined criteria) were higher than those reported in the Japanese study (26.9%, 13.2%, and 33.5%, respectively) [17], and in US adults from the National Health and Nutrition Examination Survey (26.2%, 14.2%, and 32.2%, respectively) [27]. It is possible that the prevalence rates for prediabetes and diabetes are really higher in Koreans. Alternatively, these differences in prevalence may stem from the differences in optimal cut-off values for prediabetes between ethnic groups. However, the latter possibility is less likely because ROC curve analysis showed that the optimal cut-offs for predicting progression to diabetes are 5.7 mmol/l for FPG and 5.7% (39 mmol/mol) for HbA1c, which are close to those determined in previous studies from Korea [28] and Japan [29], and are also consistent with the cut-off values recommended by the ADA.

Although FPG is slightly more sensitive and HbA1c is slightly more specific for prediction of diabetes, the overall sensitivity and specificity are similar for both methods across the clinically relevant section of the ROC curve. This may raise a question whether it is worth doing both tests since HbA1c test is more expensive. However, as we and others previously reported, there are significant discordances between the two tests; i.e., they identify different groups of people with prediabetes [12,18,27]. HbA1c can detect many subjects with prediabetes who can be missed by screening using FPG only. Therefore, combining the two tests can markedly increase the sensitivity although the specificity was decreased. However, additional studies with cost-effectiveness analysis would be needed.

Determining the actual rate of progression from prediabetes to diabetes is crucial for screening strategies. While

Table 2 – Incidence and odds ratios (ORs) of diabetes after 5 years according to baseline fasting plasma glucose and HbA1c levels and diagnosis of prediabetes.

	n	Incident diabetes (%)	Age- and sex-adjusted OR (95% CI)	Multivariate adjusted OR (95% CI)
Fasting plasma glucose				
<5.6 mmol/l	12,474	151 (1.3%)	1 (reference)	1 (reference)
5.6–6.0 mmol/l	4062	215 (5.3%)	4.27 (3.45–5.29)	3.66 (2.91–4.60)
6.1–6.9 mmol/l	1433	357 (24.9%)	24.9 (20.2–30.5)	21.1 (16.8–26.3)
HbA1c				
<5.7% (39 mmol/mol)	14,305	185 (1.3%)	1 (reference)	1 (reference)
5.7-5.9% (39–41 mmol/mol)	2561	216 (8.4%)	6.85 (5.59–8.40)	5.54 (4.47–6.88)
6.0-6.4% (42–46 mmol/mol)	1103	322 (29.2%)	30.7 (25.2–37.5)	23.2 (18.7–28.7)
Prediabetes diagnosis by				
None (normal)	10,761	48 (0.4%)	1 (reference)	1 (reference)
FPG only	3544	137 (3.9%)	8.52 (6.10–11.9)	7.66 (5.39–10.9)
HbA1c only	1713	130 (6.0%)	14.4 (10.1–20.4)	11.9 (8.24–17.3)
Both (FPG & HbA1c)	1951	435 (22.3%)	61.3 (45.0–83.4)	46.7(33.6–64.9)

FPG, fasting plasma glucose.
Prediabetes by FPG: FPG 5.6–6.9 mmol/l.
Prediabetes by HbA1c: HbA1c 5.7–6.4% (39–46 mmol/mol).
Multivariate adjusted OR: adjusted for age, sex, BMI, smoking, alcohol intake, physical activity, family history of diabetes, hypertension, serum triglycerides and HDL-cholesterol levels.

the prevalence of prediabetes was higher in our participants, the rate of progression to diabetes was lower than that in the Japanese study. The actual 5-year rates of progression in this study were 3.9% from prediabetes diagnosed by FPG criteria alone, 6.0% by HbA1c criteria alone, and 22.3% by both criteria, compared with 8.5%, 7.3%, and 37.6%, respectively, in the Japanese study [16]. The 7-year rate of progression to diabetes in American adults aged 70–79 years with IFG alone, elevated HbA1c alone, and both IFG and elevated HbA1c were 10.6%, 21.3%, and 47.9%, respectively [18]. Since annual data for FPG and HbA1c were unavailable in our study, the incidence of diabetes may have been underestimated. While these differences in the rate of progression from prediabetes to diabetes may be due to ethnic differences, they may also be due to differences in age of the participants and duration of follow up.

The percentage of people with normal HbA1c who develop diabetes is reported to be low, suggesting that these people may not require rescreening in 3 years [30]. Our results showed, however, that nearly half (49.2%) of the prediabetic individuals would have been missed if assessed only by HbA1c tests, with 3.9% of these people progressing to diabetes in 5 years. By contrast, the 5-year rate of progression to diabetes was only 0.4% in individuals with both normal FPG and normal HbA1c, compared with 22.3% in those with both increased FPG and elevated HbA1c. Therefore, individuals having normal FPG and normal HbA1c may not require rescreening in 3 years, whereas people having both elevated HbA1c and FPG may require more careful follow up and possibly aggressive treatment to reduce the risk of diabetes. The combination of FPG and HbA1c may be useful in periodic screening for diabetes, since the combination has a fairly high predictive value and both tests can be performed together by one-time blood sampling and much easier to perform than oral glucose tolerance test.

This study also confirmed that individuals with higher FPG (6.1–6.9 mmol/l) and HbA1c (6.0–6.4% [42–46 mmol/mol]) within the prediabetes group had substantially higher ORs for progression to diabetes than those with lower FPG (5.6–6.0 mmol/l) and HbA1c (5.7–5.9% [39–41 mmol/mol]), respectively. A recent meta-analysis showed that the pooled incidence rate of diabetes for HbA1c 6.0–6.4% was most similar to that for ADA-defined IFG (5.6–6.9 mmol/l) and was non-significantly lower than WHO-defined IFG (6.1–6.9 mmol/l), IGT, and IFG plus IGT [31]. In contrast, the risk of progression for HbA1c 6.0–6.4% was higher than that for FPG 5.6–6.9 mmol/l and similar to that for WHO-defined IFG (6.1–6.9 mmol/l) in our cohort. A study from Japanese cohort reported similar results with ours [17]. The reason for such discrepancy is not clear at present, but it may be attributable to racial or ethnic differences.

The cumulative incidence rate of diabetes at 5.6 years in Japanese individuals with both FPG 6.1–6.9 mmol/l and HbA1c 6.0–6.4% [42–46 mmol/mol] was 100% [17]. Although such individuals in our cohort had the highest risk for diabetes, about 50% did not progress to diabetes within 5 years. Progression to diabetes may be prevented by early implementation of effective interventions. Prevention has been repeatedly shown in studies with participants having IGT [32–34]. However, not much data are available for patients with prediabetes diagnosed by other means such as FPG or HbA1c. Prospective controlled trials are needed to confirm the effectiveness of preventive interventions in these groups.

As reported previously, prediabetic individuals who met HbA1c criteria alone were significantly older and more likely to be women, whereas those who met FPG criteria alone had higher BMI, waist circumference, and systolic and diastolic blood pressure [14]. In addition, HbA1c levels have been positively associated with age independent of glucose levels [35], with the specificity of HbA1c-based diagnostic criteria

for prediabetes decreasing substantially with increasing age [36]. Obesity also influenced the ability of HbA1c to diagnose prediabetes and diabetes in an Italian cohort [11]. However, in our study, the differences in the ability of FPG and HbA1c to predict progression to diabetes were consistent over all subgroups of age, sex, BMI, and family history of diabetes, with no significant interactions with these factors.

Our study had several potential limitations. First, our study population was comprised of individuals who came in for voluntary medical check-ups and is therefore not representative of the general population. Second, the lack of oral glucose tolerance test results may have led to the inclusion of participants with undiagnosed diabetes at baseline and under-diagnosis of incident diabetes at follow up. Third, diabetes status was determined at only two time points, at baseline and follow up. As diabetic status was not determined between these time points, we could not know the exact time interval between baseline and development of diabetes and the incidence of diabetes may have been underestimated. Fourth, the ascertainment of diabetes was made by a single measurement of FPG and HbA1c, which differs from the accepted WHO clinical definition which requires confirmation test in the absence of symptoms. This will increase the incidence, and likely decrease the AUC, as the specificity of diabetes diagnosis was less than optimal. Despite these limitations, however, the large sample size and the diverse age, sex, and socio-economic status of the participants enabled an estimation of actual progression rate in a real world setting.

In conclusion, although HbA1c criteria identified fewer individuals with prediabetes than FPG, the ability of HbA1c to predict progression to diabetes was higher than that of FPG. However, these two tests could independently and additively predict risk of progression to diabetes, with the combination of the two tests more efficient at identifying individuals at risk for diabetes. Further prospective controlled trials are needed to determine whether early screening for prediabetes, using a combination of HbA1c and FPG, can be beneficial in preventing or delaying the development of type 2 diabetes and its complications.

Conflict of interests

The authors have no potential conflicts of interest to declare in relation to this study.

Funding sources

None.

REFERENCES

[1] Hollander P, Spellman C. Controversies in prediabetes: do we have a diagnosis? Postgrad Med 2012;124:109–18.
[2] American Diabetes Association. Diagnosis and classification of diabetes mellitus. Diabetes Care 2010;33(Suppl 1):S62–9.
[3] World Health Organization. Definition, diagnosis and classification of diabetes mellitus and its complications. Part 1: diagnosis and classification of diabetes mellitus. Geneva: World Health Organization; 1999.
[4] International Expert Committee. International Expert Committee report on the role of the A1C assay in the diagnosis of diabetes. Diabetes Care 2009;32:1327–34.
[5] Mann DM, Carson AP, Shimbo D, Fonseca V, Fox CS, Muntner P. Impact of A1C screening criterion on the diagnosis of prediabetes among U.S. adults. Diabetes Care 2010;33:2190–5.
[6] Cowie CC, Rust KF, Byrd-Holt DD, Gregg EW, Ford ES, Geiss LS, et al. Prevalence of diabetes and high risk for diabetes using A1C criteria in the U.S. population in 1988–2006. Diabetes Care 2010;33:562–8.
[7] Bhansali A, Walia R, Ravi Kumar P, Ravi Kiran M, Shanmugasundar G. Accuracy of glycated haemoglobin in screening for pre-diabetes in Asian Indians – a community survey: the Chandigarh Urban Diabetes Study (CUDS). Diabet Med 2012;29:1385–9.
[8] Herman WH, Ma Y, Uwaifo G, Haffner S, Kahn SE, Horton ES, et al. Differences in A1C by race and ethnicity among patients with impaired glucose tolerance in the Diabetes Prevention Program. Diabetes Care 2007;30:2453–7.
[9] Herman WH, Dungan KM, Wolffenbuttel BH, Buse JB, Fahrbach JL, Jiang H, et al. Racial and ethnic differences in mean plasma glucose, hemoglobin A1c, and 1,5-anhydroglucitol in over 2000 patients with type 2 diabetes. J Clin Endocrinol Metab 2009;94:1689–94.
[10] Likhari T, Gama R. Glycaemia-independent ethnic differences in HbA(1c) in subjects with impaired glucose tolerance. Diabet Med 2009;26:1068–9.
[11] Incani M, Sentinelli F, Perra L, Pani MG, Porcu M, Lenzi A, et al. Glycated hemoglobin for the diagnosis of diabetes and prediabetes: diagnostic impact on obese and lean subjects, and phenotypic characterization. J Diabetes Invest 2015;6:44–50.
[12] Kim CH, Kim HK, Bae SJ, Park JY, Lee KU. Discordance between fasting glucose-based and hemoglobin A1c-based diagnosis of diabetes mellitus in Koreans. Diabetes Res Clin Pract 2011;91:e8–e10.
[13] Kim HK, Bae SJ, Choe J. Impact of HbA1c criterion on the detection of subjects with increased risk for diabetes among health check-up recipients in Korea. Diabetes Metab J 2012;36:151–6.
[14] Kim CH, Kim HK, Kim BY, Jung CH, Mok JO, Kang SK. Impact of hemoglobin A1c-based criterion on diagnosis of prediabetes: the Korea National Health and Nutrition Examination Survey 2011. J Diabetes Invest 2015;6:51–5.
[15] Cederberg H, Saukkonen T, Laakso M, Jokelainen J, Harkonen P, Timonen M, et al. Postchallenge glucose, A1C, and fasting glucose as predictors of type 2 diabetes and cardiovascular disease: a 10-year prospective cohort study. Diabetes Care 2010;33:2077–83.
[16] Heianza Y, Hara S, Arase Y, Saito K, Fujiwara K, Tsuji H, et al. HbA1c 5.7–6.4% and impaired fasting plasma glucose for diagnosis of prediabetes and risk of progression to diabetes in Japan (TOPICS 3): a longitudinal cohort study. Lancet 2011;378:147–55.
[17] Heianza Y, Arase Y, Fujihara K, Tsuji H, Saito K, Hsieh SD, et al. Screening for pre-diabetes to predict future diabetes using various cut-off points for HbA(1c) and impaired fasting glucose: the Toranomon Hospital Health Management Center Study 4 (TOPICS 4). Diabet Med 2012;29:e279–85.
[18] Lipska KJ, Inzucchi SE, Van Ness PH, Gill TM, Kanaya A, Strotmeyer ES, et al. Elevated HbA1c and fasting plasma glucose in predicting diabetes incidence among older adults: are two better than one? Diabetes Care 2013;36:3923–9.
[19] Admiraal WM, Holleman F, Snijder MB, Peters RJ, Brewster LM, Hoekstra JB, et al. Ethnic disparities in the association of impaired fasting glucose with the 10-year cumulative

incidence of type 2 diabetes. Diabetes Res Clin Pract 2014;103:127–32.
[20] DeLong ER, DeLong DM, Clarke-Pearson DL. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. Biometrics 1988;44:837–45.
[21] Sato KK, Hayashi T, Harita N, Yoneda T, Nakamura Y, Endo G, et al. Combined measurement of fasting plasma glucose and A1C is effective for the prediction of type 2 diabetes: the Kansai Healthcare Study. Diabetes Care 2009;32:644–6.
[22] Hu Y, Liu W, Chen Y, Zhang M, Wang L, Zhou H, et al. Combined use of fasting plasma glucose and glycated hemoglobin A1c in the screening of diabetes and impaired glucose tolerance. Acta Diabetol 2010;47:231–6.
[23] Law LS, Tso AW, Tam S, Wat NM, Cheung BM, Lam KS. Haemoglobin A1c is superior to fasting glucose in predicting the incidence of diabetes over 8 years among Chinese. Diabetes Res Clin Pract 2011;91:e53–6.
[24] Nakagami T, Tajima N, Oizumi T, Karasawa S, Wada K, Kameda W, et al. Hemoglobin A1c in predicting progression to diabetes. Diabetes Res Clin Pract 2010;87:126–31.
[25] Valdes S, Botas P, Delgado E, Alvarez F, Diaz-Cadorniga F. HbA (1c) in the prediction of type 2 diabetes compared with fasting and 2-h post-challenge plasma glucose: the Asturias study (1998–2005). Diabetes Metab 2011;37:27–32.
[26] Zhang X, Gregg EW, Williamson DF, Barker LE, Thomas W, Bullard KM, et al. A1C level and future risk of diabetes: a systematic review. Diabetes Care 2010;33:1665–73.
[27] James C, Bullard KM, Rolka DB, Geiss LS, Williams DE, Cowie CC, et al. Implications of alternative definitions of prediabetes for prevalence in U.S. adults. Diabetes Care 2011;34:387–91.
[28] Bae JC, Rhee EJ, Lee WY, Park SE, Park CY, Oh KW, et al. Optimal range of HbA1c for the prediction of future diabetes: a 4-year longitudinal study. Diabetes Res Clin Pract 2011;93:255–9.
[29] Kato M, Noda M, Suga H, Nakamura T, Matsumoto M, Kanazawa Y, et al. Haemoglobin A1c cut-off point to identify a high risk group of future diabetes: results from the Omiya MA Cohort Study. Diabet Med 2012;29:905–10.
[30] Edelman D, Olsen MK, Dudley TK, Harris AC, Oddone EZ. Utility of hemoglobin A1c in predicting diabetes risk. J Gen Intern Med 2004;19:1175–80.
[31] Morris DH, Khunti K, Achana F, Srinivasan B, Gray LJ, Davies MJ, et al. Progression rates from HbA1c 6.0-6.4% and other prediabetes definitions to type 2 diabetes: a meta-analysis. Diabetologia 2013;56:1489–93.
[32] Pan XR, Li GW, Hu YH, Wang JX, Yang WY, An ZX, et al. Effects of diet and exercise in preventing NIDDM in people with impaired glucose tolerance. The Da Qing IGT and Diabetes Study. Diabetes Care 1997;20:537–44.
[33] Tuomilehto J, Lindstrom J, Eriksson JG, Valle TT, Hamalainen H, Ilanne-Parikka P, et al. Prevention of type 2 diabetes mellitus by changes in lifestyle among subjects with impaired glucose tolerance. N Engl J Med 2001;344:1343–50.
[34] Ramachandran A, Snehalatha C, Mary S, Mukesh B, Bhaskar AD, Vijay V. The Indian Diabetes Prevention Programme shows that lifestyle modification and metformin prevent type 2 diabetes in Asian Indian subjects with impaired glucose tolerance (IDPP-1). Diabetologia 2006;49:289–97.
[35] Pani LN, Korenda L, Meigs JB, Driver C, Chamany S, Fox CS, et al. Effect of aging on A1C levels in individuals without diabetes: evidence from the Framingham Offspring Study and the National Health and Nutrition Examination Survey 2001–2004. Diabetes Care 2008;31:1991–6.
[36] Dubowitz N, Xue W, Long Q, Ownby JG, Olson DE, Barb D, et al. Aging is associated with increased HbA1c levels, independently of glucose levels and insulin resistance, and also with decreased HbA1c diagnostic specificity. Diabet Med 2014;31:927–35.